MILITARY STANDARD

MIL D-1320-49A
(NAVY)
26 MAY 1982

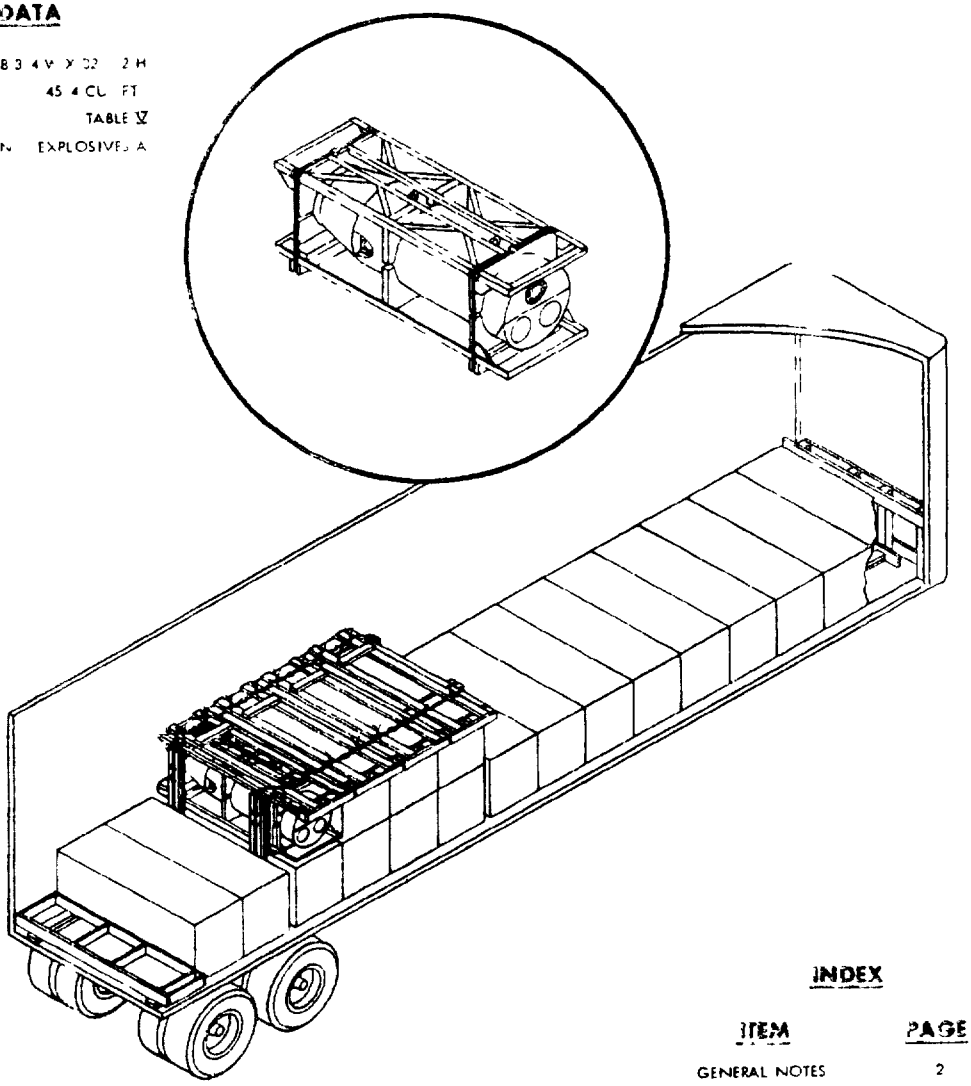
SUPERSEDING
MIL-STD-1320-49
28 AUGUST 1978

TRUCKLOADING

MINE, UNDERWATER MK 55 AND MODS
IN CRATE UNDERWATER MINE MK 55 MOD 1
DOMESTIC UNIT LOAD WR-53/714

UNIT LOAD DATA

DIMENSIONS	84 L X 28 3/4 W X 22 2/3 H
CUBE	45.4 CU FT
GROSS WEIGHT	TABLE V
DOT HAZARD CLASSIFICATION	EXPLOSIVES A



INDEX

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UNITIZING	9

NOTES

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY)

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENT ONLY

SIGNATURE TECHNICAL DIRECTION AGENT (TDA) — DATE 4/21/82

SIGNATURE SEASYSCOM TECHNICAL DIRECTION — DATE 5/24/82

ORIGINATOR — DATE 12 APR 1982

NAVAL WEAPONS HANDLING CENTER
WPNSTA EARLE, NEW JERSEY

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GENERAL NOTES

1 THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING DOMESTIC UNIT LOADS OF THE UNDERWATER MINE MK 55 AND MODS IN CRATE UNDERWATER MINE MK 55 MOD 1 THE CRATED MINE SHALL HAVE SKIDS IN ACCORDANCE WITH WR-53 714

2 THE PROCEDURES AND PRACTICES HEREIN ARE INTENDED FOR 40, 42 44 AND 45 FT VAN TYPE TRAILERS WITH NAILABLE FLOORS TRAILERS SHALL HAVE THEIR AXLES LOCATED IN THE "WESTERN POSITION (AT THE EXTREME REAR OF THE TRAILER) DO NOT USE TRAILERS WITH ROLL UP DOORS TRACTOR SHOULD HAVE SLIDING 5TH WHEEL

3 THE LENGTH OF THE STRUT PIECE 5 OF DETAIL C (PAGE 5) SHALL BE OF SUFFICIENT LENGTH TO CAUSE THE BLOCKING ASSEMBLY TO FILL THE VOID BETWEEN THE LADING AND THE TRAILER DOORS WHEN THE DOORS ARE CLOSED

4 THE LENGTH OF THE STRUT, PIECE 7 OF DETAIL D (PAGE 5) SHALL BE OF SUFFICIENT LENGTH TO CAUSE THE SWAY BRACE TO FILL THE VOID BETWEEN THE LADING AND THE SIDE WALL OF THE TRAILER TWIST-TIE THE SWAY BRACE ASSEMBLY TO THE TOP OF MINE CRATE (1 PLACES) WITH 16 GAUGE SOFT ANNEALED IRON WIRE

5 DUE TO THE OVERHANG OF THE MINE CRATE SLEEPERS SHALL BE PREPOSITIONED 74 INCHES FROM THE LEFT SIDE OF THE TRAILER AS SHOWN IN THIS DRAWING TAIL OF THE MINE SHALL BE AGAINST THE LEFT SIDE WALL OF THE TRAILER

6 WHEN THE SPACE BETWEEN THE LADING AND THE REAR OF THE TRAILER IS GREATER THAN 36 INCHES BLOCK THE REAR OF THE LOAD AS SHOWN ON PAGE 8

7 THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS

8 AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO INSIDE OF TRAILER IN AN ACCESSIBLE AREA CLOSE AND SEAL TRAILER DOORS AND ATTACH EXPLOSIVES "A" PLACARD TO BOTH SIDES FRONT AND BACK OF TRAILER

9 APPLICABLE MATERIAL SPECIFICATIONS DUNNAGE LUMBER MM-L-751 NAILS FF-N-105 TYPE II STYLE 10 COMMON BRIGHT, STRAPPING QQ-S-781 TYPE I FINISH A B OR C

10 FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY)

FTL & LTL 40,42,44 & 45 FT TRAILERS (VAN)

WARNING

THIS STANDARD PROVIDES FOR INCREASED LOADINGS WHICH PRODUCE HEAVIER GROSS TANDEM AXLE WEIGHTS AND GROSS VEHICLE WEIGHTS THESE INCREASED WEIGHTS ARE NOT AUTHORIZED IN ALL 50 STATES

1 TABLES I II III AND IV PROVIDE THE CORRECT LOAD PATTERN NUMBER TO BE USED WHEN SHIPPING THE UNDERWATER MINE MK 56 AND MODS TO USE THESE TABLES -

A DETERMINE THE DODIC NALC NUMBER OF THE MINE

B SELECT THE TABLE LISTING THIS NUMBER

C DETERMINE THE TYPE OF TRACTOR-TRAILER (LIGHTWEIGHT OR STANDARD WEIGHT) TO BE LOADED

NOTE

LIGHT WEIGHT TRACTOR-TRAILER - TARE WEIGHT OF TRAILER TANDEM AXLES 6 800 LBS MAXIMUM AND THE DRIVE TANDEM AXLES 10 500 LBS MAXIMUM

STANDARD WEIGHT TRACTOR-TRAILER - TARE WEIGHT OF TRAILER TANDEM AXLES 7,800 LBS MAXIMUM AND THE DRIVE TANDEM AXLES 11 500 LBS MAXIMUM

THE ABOVE MAXIMUM WEIGHTS MAY BE ADJUSTED WHEN EXPERIENCE INDICATES THAT THE WEIGHT RESTRICTIONS CAN BE MET

D DETERMINE IF THE ROUTING IS THROUGH STATES PERMITTING 32 000 LBS OR 34 000 LBS ON TANDEM AXLES

L USING THE COLUMN FOR THE LENGTH OF THE TRAILER BEING LOADED FIND THE LOAD PATTERN NUMBER TO BE USED FOR THE NUMBER OF ITEMS TO BE LOADED

2 TABLE VI PROVIDES THE PRESCRIBED LOAD PATTERNS AFTER DETERMINING THE CORRECT LOAD PATTERN NUMBER (SEE NOTE 1 ABOVE), FIND THE CORRECT LOAD PATTERN TO USE OPPOSITE THE LOAD PATTERN NUMBER

3 BLOCK AND BRACE ALL LOAD PATTERNS SHOWN IN TABLE VI USING THE PRINCIPLES SHOWN ON PAGES 6 AND 7

4 WHEN TRUCKLOAD HAS A PARTIAL SECOND LAYER THE CRATED MINE(S) IN THE SECOND LAYER SHALL BE STRAPPED TO THE BOTTOM CRATED MINE(S) WHERE THE LOAD PATTERN CHANGES FROM TWO LAYERS HIGH TO ONE LAYER HIGH TWO HIGH CRATES AT THE REAR OF THE LOAD SHALL ALSO BE STRAPPED TOGETHER SEE UNITIZING PROCEDURE PAGE 9 FOR INSTRUCTIONS

5 WHEN THE CRATED MINES ARE TWO HIGH AT THE REAR OF THE LOAD, THE LAST TWO STACKS SHALL BE STRAPPED TOGETHER WITH TWO 1 1 4 X .035 STEEL STRAPS THESE STRAPS ENCIRCLE THE TWO STACKS PASSING OVER THE TOP TWO CRATED MINES AND UNDER THE BOTTOM TWO TENSION STRAPS AND SEAL WITH TWO DOUBLE CRIMPED SEALS SEE PAGE 6 THESE MINES SHOULD BE STRAPPED TOGETHER IN PLACE ATTACH TO EACH STRAP IN A CONSPICUOUS PLACE A SUITABLE TAG CARRYING THE FOLLOWING INSTRUCTION CUT AND REMOVE THIS STRAP BEFORE UNLOADING VEHICLE

WHEN SHIPPING LESS THAN A FULL SINGLE LAYER FOLLOW THE PROCEDURES SHOWN ON PAGE 8

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TABLE I

UNDERWATER MINE MK 55 AND MODS
DODIC/NALC NUMBER
U923, U929, U935, U936 AND U937

NO OF UNIT LOADS	LOAD PATTERN NUMBER											
	LIGHTWEIGHT TRACTOR-TRAILER						STANDARD WEIGHT TRACTOR-TRAILER					
	MAX AXLE WT 34 000 LBS			MAX AXLE WT 32 000 LBS			MAX AXLE WT 34 000 LBS			MAX AXLE WT 32 000 LBS		
	TRAILER LENGTH			TRAILER LENGTH			TRAILER LENGTH			TRAILER LENGTH		
	40	42	44/45	40	42	44/45	40	42	44/45	40	42	44/45
20	22	13	5	NA	NA	NA	NA	NA	NA	NA	NA	NA
19	23	14	6	23	14	6	23	14	6	NA	NA	NA
18	24	15	7	24	15	7	24	15	7	24	15	7
17	25	16	8	25	16	8	25	16	8	25	16	8
16	26	17	*	26	17	*	26	17	*	26	17	*
15	27	*	*	27	*	*	27	*	*	27	*	*
14	*	*	*	*	*	*	*	*	*	*	*	*

TABLE II

UNDERWATER MINE MK 55 AND MODS
DODIC NALC NUMBER
U887, U888, U889, U899, U900, U901, U924, U925, U930 AND U931

NO OF UNIT LOADS	LOAD PATTERN NUMBER											
	LIGHTWEIGHT TRACTOR-TRAILER						STANDARD WEIGHT TRACTOR-TRAILER					
	MAX AXLE WT 34 000 LBS			MAX AXLE WT 32 000 LBS			MAX AXLE WT 34 000 LBS			MAX AXLE WT 32 000 LBS		
	TRAILER LENGTH			TRAILER LENGTH			TRAILER LENGTH			TRAILER LENGTH		
	40	42	44/45	40	42	44/45	40	42	44/45	40	42	44/45
21	21	12	4	NA	NA	NA	NA	NA	NA	NA	NA	NA
20	22	13	5	22	13	5	22	13	5	NA	NA	NA
19	23	14	6	23	14	6	23	14	6	23	14	6
18	24	15	7	24	15	7	24	15	7	24	15	7
17	25	16	8	25	16	8	25	16	8	25	16	8
16	26	17	*	26	17	*	26	17	*	26	17	*
15	27	*	*	27	*	*	27	*	*	27	*	*
14	*	*	*	*	*	*	*	*	*	*	*	*

TABLE III

UNDERWATER MINE MK 55 AND MODS
DODIC/NALC NUMBER
U875, U876, U877 U911 U912 AND U913

NO OF UNIT LOADS	LOAD PATTERN NUMBER											
	LIGHTWEIGHT TRACTOR-TRAILER						STANDARD WEIGHT TRACTOR-TRAILER					
	MAX AXLE WT 34 000 LBS			MAX AXLE WT 32,000 LBS			MAX AXLE WT, 34 000 LBS			MAX AXLE WT 32 000 LBS		
	TRAILER LENGTH			TRAILER LENGTH			TRAILER LENGTH			TRAILER LENGTH		
	40	42	44/45	40	42	44/45	40	42	44/45	40	42	44/45
22	20	11	3	NA	NA	NA	NA	NA	NA	NA	NA	NA
21	21	12	4	21	12	4	21	12	4	NA	NA	NA
20	22	13	5	22	13	5	22	13	5	22	13	5
19	23	14	6	23	14	6	23	14	6	23	14	6
18	24	15	7	24	15	7	24	15	7	24	15	7
17	25	16	8	25	16	8	25	16	8	25	16	8
16	26	17	*	26	17	*	26	17	*	26	17	*
15	27	*	*	27	*	*	27	*	*	27	*	*
14	*	*	*	*	*	*	*	*	*	*	*	*

*WHEN LOADING THIS QUANTITY OR LESS USE A ONE HIGH SINGLE ROW PATTERN BLOCKED AND BRACED USING THE PRINCIPLES SHOWN ON PAGE 4

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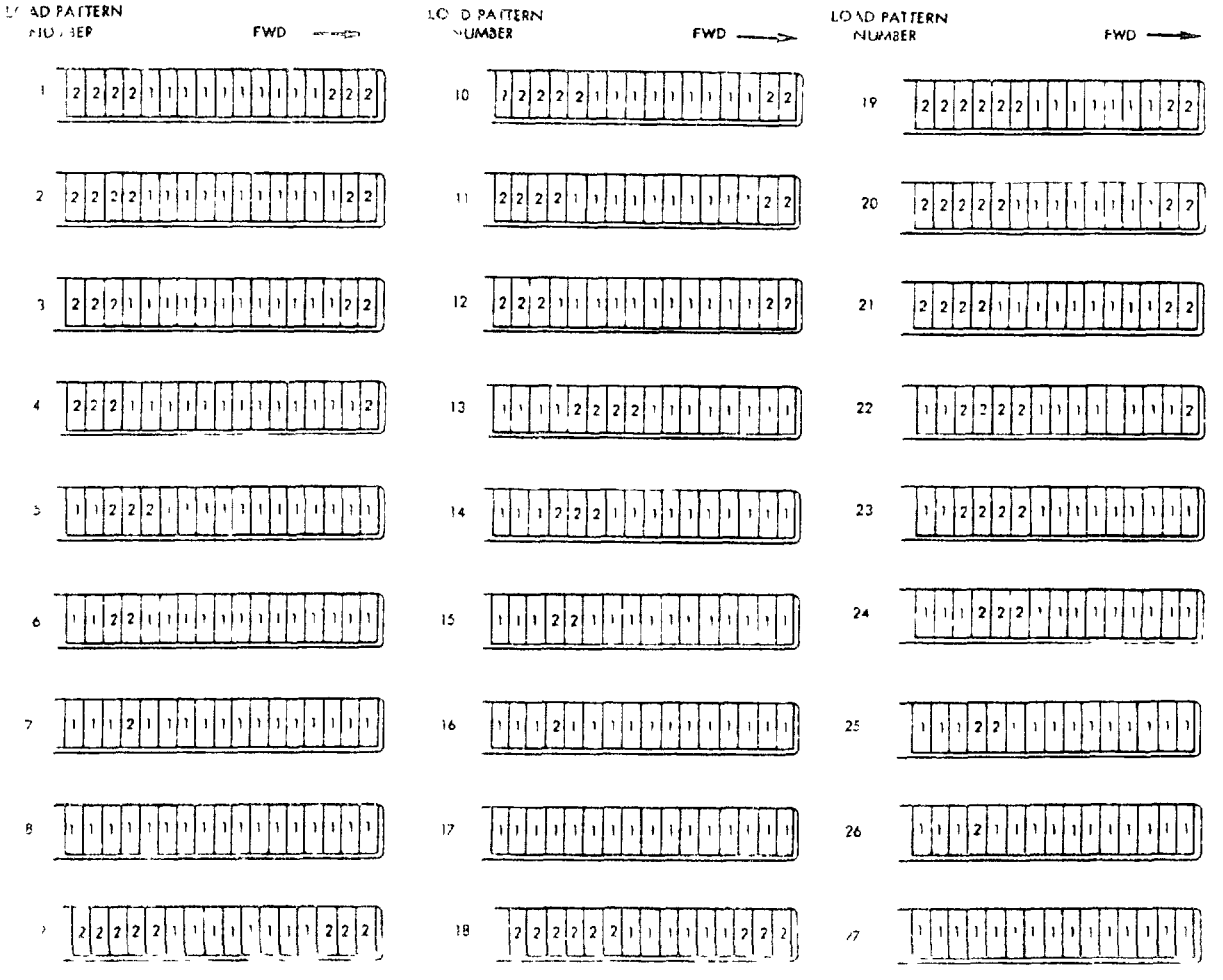
TABLE IV

UNIT LOAD DATA

MODIFY / DALC NO	MINE MK 55 MOD	CONFIG URATION	GROSS WEIGHT (APPROX.) LBS CRATED MINE	GROSS WEIGHT (APPROX.) LBS UNIT LOAD
U875	1	B	2185	2206
U876	1	C	2163	2184
U877	1	D	2155	2176
U887	2	B	2269	2284
U888	2	C	2229	2250
U889	2	D	2221	2242
U899	3	B	2276	2297
U900	3	C	2279	2260
U901	3	D	2231	2252
U911	4	B	2168	2189
U912	4	C	2160	2181
U913	4	D	2135	2156
U923	5	B	2422	2443
U924	5	C	2236	225
U925	5	D	2232	2253
U929	6	B	2380	2401
U930	6	C	2247	2268
U931	6	D	2232	2253
U935	7	B	2325	2346
U936	7	C	2315	2336
U937	7	D	2285	2306

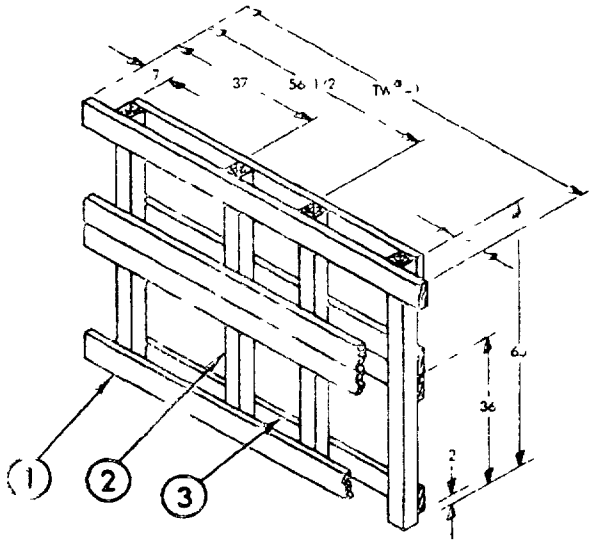
TABLE V

LOAD PATTERN



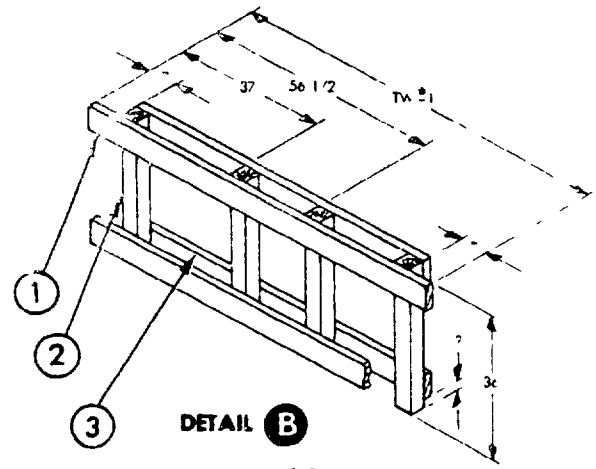
NOTE NUMBER IN BLOCKS INDICATE NUMBER OF CRATED MINES IN A LAYER.

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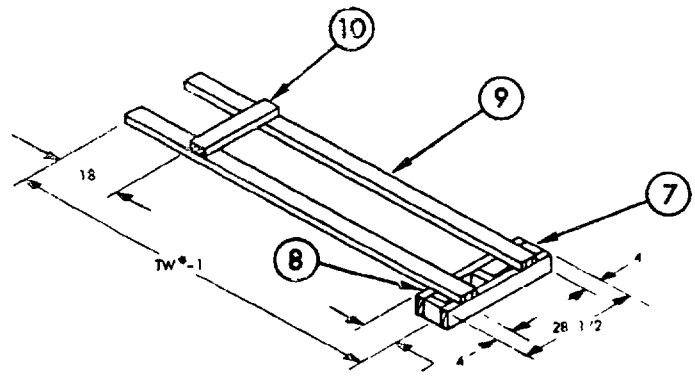
DETAIL A

FORWARD BULKHEAD
(1 REQUIRED)



DETAIL B

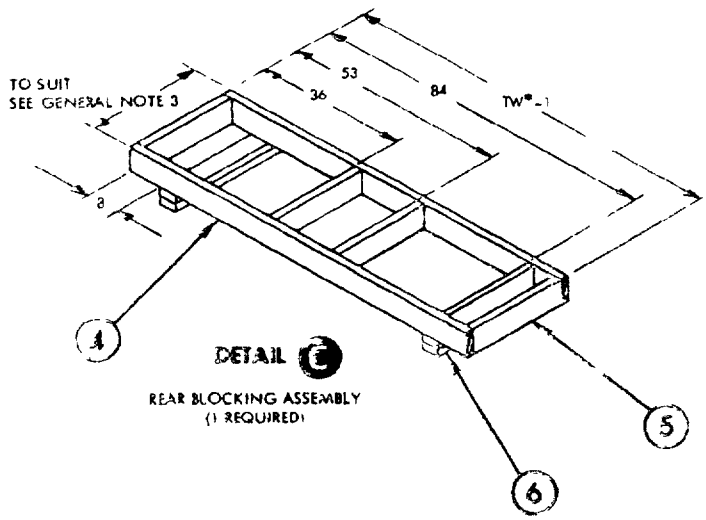
FORWARD BULKHEAD
(1 REQUIRED)



TO SUIT
SEE GENERAL NOTE 4

DETAIL D

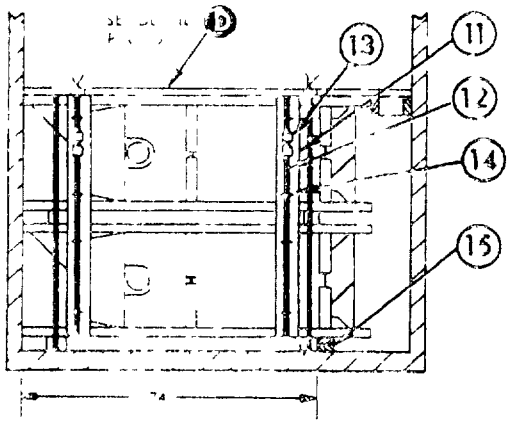
SWAY BRACE ASSEMBLY



DETAIL C

REAR BLOCKING ASSEMBLY
(1 REQUIRED)

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SECTION E-E

NAIL 1ST PIECE TO DECK W 1 - 16d NAIL EVERY 8 INCHES NAIL 2ND PIECE TO 1ST IN LIKE MANNER

SEE DETAIL D PAGE 5

SEE DETAIL A PAGE 5

ARROW INDICATES TAIL END OF MINES

TWIST TIE TO STEEL BANDING WITH 16 GA SOFT ANNEALED IRON WIRE 4 PLACES EACH STACK

SEE DETAIL C PAGE 5

SEE FTL NOTE 5

2 X 6 X 14 DOUBLED LAMINATED TC DETAIL C 10d NAILS EACH PIECE

*TW TRAILER WIDTH

PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ D	NAIL TO	NUMBER NAILS	SIZE
18	SHIM	2 x 6 X 14	2	SEE FIELD NOTE		
17	SEAL	FOR 1 1/4 STRAP	4			
16	STRAP	1 1/4 X .035 X 22 FT	2			
15	SLEEPER	2 X 4 X RANDOM LENGTHS	144 FT	SEE SECTION E-E		
14	STAPLE	FOR 1 1/4 STRAP	24	-	-	-
13	SEAL	FOR 1 1/4 STRAP	12	-	-	-
12	STRAP	1 1/4 X .035 X 12 FT.	6	-	-	-
11	STIFFENER	2 X 6 X 61 1/2	12	LAMINATE	6	10d
10	CLEAT	2 X 4 X 20 1/2	9	9	2/JOINT CLINCHED	16d

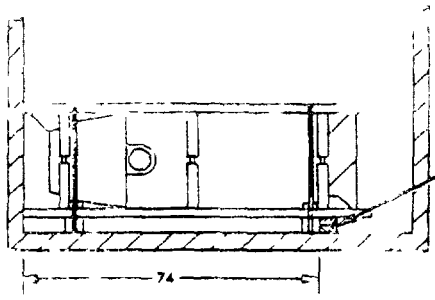
LIST OF MATERIALS & NAILING DATA

PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ D	NAIL TO	NUMBER NAILS	SIZE
9	SUPPORT	2 X 4 X TW*-1	18	8	2/JOINT	16d
8	LONGITUDINAL	2 X 4 X 28 1/2	18	7	2/JOINT	12d
7	STRUT	2 X 4 TO SUIT	27	SEE 8	-	-
6	SUPPORT	2 X 4 TO SUIT	4	4	2/JOINT	10d
				6	3/PIECE	10d
5	STRUT	2 X 6 TO SUIT	5	SEE 4	-	-
4	CROSSMEMBER	2 X 6 X TW*-1	2	5	3/JOINT	16d
3	FWD CROSSMEMBER	2 X 6 TO SUIT	4	2	3/JOINT	12d
2	VERTICAL	4 X 4 X 65	4	SEE 1 & 3	-	-
1	AFT CROSSMEMBER	2 X 6 X TW*-1	4	2	3/JOINT	12d

LIST OF MATERIALS & NAILING DATA

40 FT VAN TRAILER WITH 24 MINES

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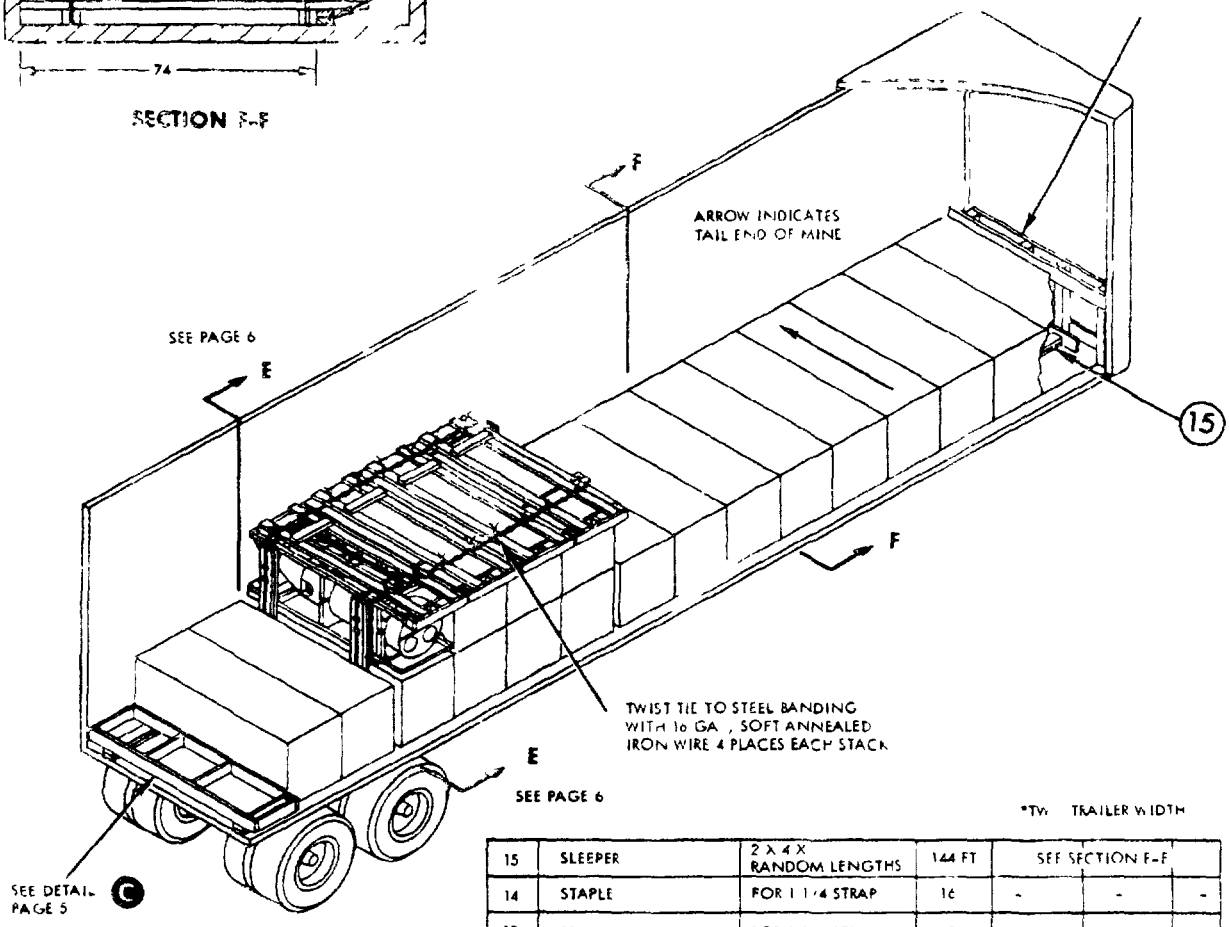


SECTION F-F

NAIL 1ST PIECE TO DECK W 1-16d NAIL EVERY 8 INCHES NAIL 2ND PIECE TO 1ST IN LIKE MANNER

15

SEE DETAIL B PAGE 5

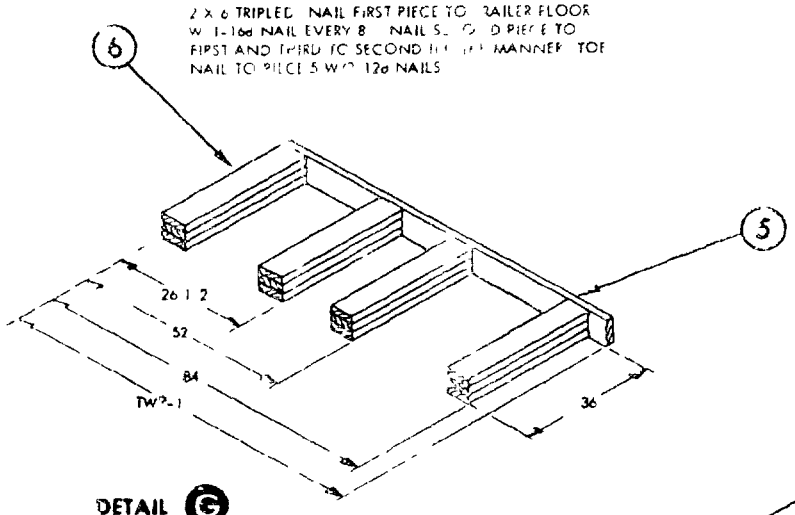


*Tw TRAILER WIDTH

PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TO	NAILS NUMBER	NAILS SIZE
15	SLEEPER	2 X 4 X RANDOM LENGTHS	144 FT	SEE SECTION F-F		
14	STAPLE	FOR 1 1/4 STRAP	16	-	-	-
13	SEAL	FOR 1 1/4 STRAP	8	-	-	-
12	STRAP	1 1/4 X .035 X 17 FT	4	-	-	-
11	STIFFENER	2 X 6 X 6 1/2	8	LAMINATE	6	10d
10	CLEAT	2 X 4 X 20 1/2	4	9	2 JOINT CLINCHED	16d
9	SUPPORT	2 X 4 X TW*-1	8	8	2 JOINT	16d
8	LONGITUDINAL	2 X 4 X 26 1/2	8	7	2 JOINT	12d
7	STRUT	2 X 4 TO SUIT	12	SEE 8	-	-
6	SUPPORT	2 X 4 TO SUIT	4	4 6	2 JOINT 3 PIECE	10d 16d
5	STRUT	2 X 6 TO SUIT	5	SEE 4	-	-
4	CROSSMEMBER	2 X 6 X TW*-1	2	5	3 JOINT	16d
3	FWD CROSSMEMBER	2 X 6 TO SUIT	2	2	3 JOINT	12d
2	VERTICAL	4 X 4 X 36	4	-	-	-
1	AFT CROSSMEMBER	2 X 6 X TW*-1	2	2	3 JOINT	12d

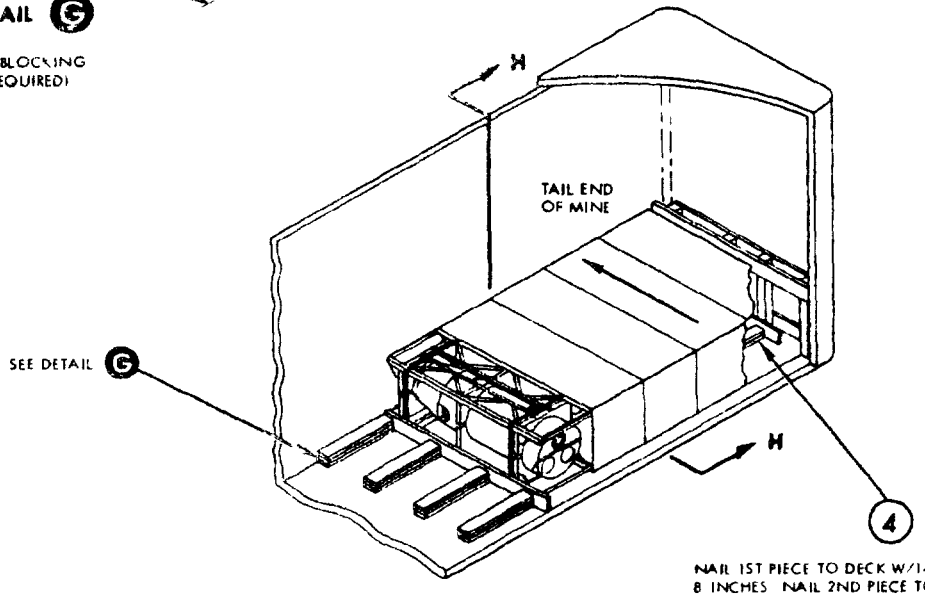
LIST OF MATERIALS & NAILING DATA

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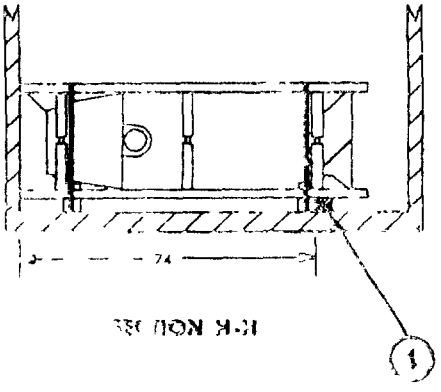


DETAIL G

REAR BLOCKING
(1 REQUIRED)



NAIL 1ST PIECE TO DECK W/1-16d NAIL EVERY 8 INCHES NAIL 2ND PIECE TO 1ST IN LIKE MANNER

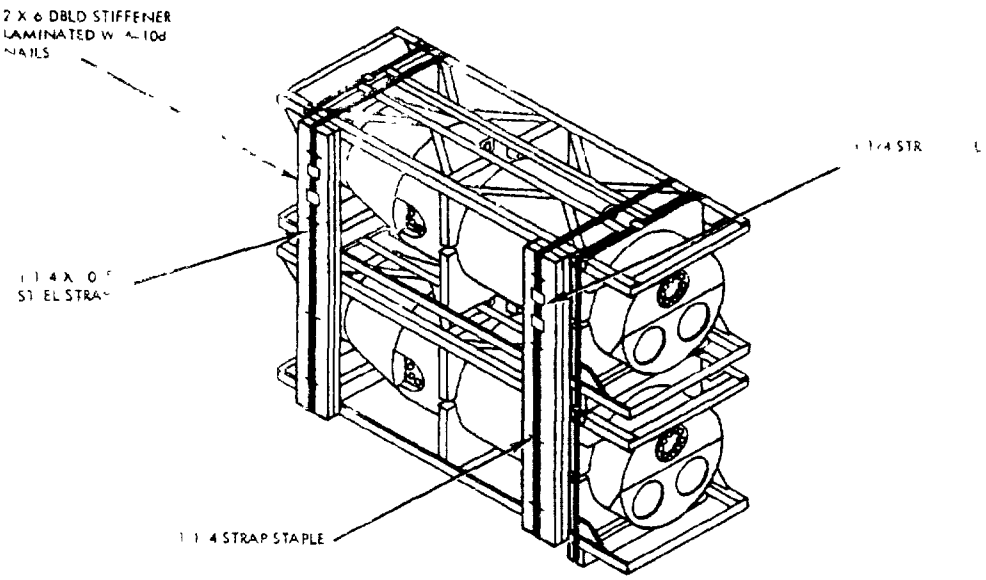


PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
	LIST OF MATERIALS & NAILING DATA				NAILS	
6	BACKUP CLEAT	2 X 6 X 36	12	SEE FIELD NOTE		
5	CROSSMEMBER	2 X 6 X TWP-1	1	SEE FIELD NOTE		
4	SLEEPER	2 X 4 TO SUIT	AS REQD	SEE FIELD NOTE		
3	PWD CROSSMEMBER	2 X 6 TO SUIT	2	2	3	12d
2	VERTICAL	4 X 4 X 36	4	SEE 1 & 3	-	-
1	AFT CROSSMEMBER	2 X 6 X TWP-1	2	2	3	12d

TYPICAL AXIAL SINGLE LAYER

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UNITIZING PROCEDURE



1. WHEN REQUIRED BY THIS MIL-STD, THE TWO HIGH STACKED UNIT LOADS SHALL BE UNITIZED

NOTE

WHEN LOADING A TRAILER WITH A SINGLE STACK OF UNIT LOADS, THE STIFFENERS SHALL BE AT BOTH SIDES

2. THE 2 X 6 DOUBLED STIFFENERS SHALL EXTEND FROM THE TOP FRAME OF THE TOP CRATE TO THE BOTTOM ANGLE OF THE BOTTOM CRATE
3. THE 1 1/4 X .035 STEEL STRAPS POSITIONED AS SHOWN ENCIRCLE THE STACKED MINE CRATES WITH THE STRAPS HOLDING THE STIFFENERS IN PLACE
4. TENSION STRAPS AND SEAL WITH TWO DOUBLE CRIMPED SEALS. SECURE EACH STRAP TO THE STIFFENER WITH FOUR 1 1/4 INCH STAPLES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-STD-1320-1DA

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

a GIVE PARAGRAPH NUMBER AND WORDING

b RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

DD FORM 1426 JAN 72

REPLACES EDITION OF JAN 68 WHICH IS OBSOLETE

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Commanding Officer
Naval Weapons Station Earle
Naval Weapons Handling Center (803)
Colts Neck, NJ 07722

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